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Deep Reinforcement Learning

HOMEWORK 2 — POLICY-BASED METHODS

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1 Policy Gradients & Variance Reduction

Question 1: The Score Function Estimator [3 Points]

To optimize $J(\theta)$, we need its gradient $\nabla_\theta J(\theta)$. However, we cannot simply push the gradient inside the expectation because the expectation itself depends on the parameters θ. Show that the gradient of the objective function can be rewritten as an expectation over trajectories:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau)\right]$$

Answer:

To derive the policy gradient for the objective function $J(\theta) = \mathbb{E}_{\times\sim\pi_\theta}[R(\tau)]$, we first express the expectation explicitly as an integral or sum over all possible trajectories τ: $J(\theta) = \sum_\tau P(\tau|\theta)R(\tau)$ Taking the gradient with respect to the policy parameters θ yields: $\nabla_\theta J(\theta) = \sum_\tau \nabla_\theta P(\tau|\theta)R(\tau)$ Using the log-derivative trick (or score function estimator), we utilize the calculus identity $\nabla_\theta \log P(\tau|\theta) = \frac{\nabla_\theta P(\tau|\theta)}{P(\tau|\theta)}$, which allows us to substitute $\nabla_\theta P(\tau|\theta) = P(\tau|\theta)\nabla_\theta \log P(\tau|\theta)$:

$$\nabla_\theta J(\theta) = \sum_\tau P(\tau|\theta)\nabla_\theta \log P(\tau|\theta)R(\tau) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\nabla_\theta \log P(\tau|\theta)R(\tau)\right]$$

The probability of a trajectory $\tau = (s_0, a_0, s_1, a_1, \ldots, s_T, a_T)$ under a parameterized policy π_θ and environment transition dynamics $P(s_{t+1}|s_t, a_t)$ is defined as:

$$P(\tau|\theta) = P(s_0)\prod_{t=0}^{T} \pi_\theta(a_t|s_t)P(s_{t+1}|s_t, a_t)$$

Applying the natural logarithm to both sides decomposes the cumulative products into a sum of log-probabilities:

$$\log P(\tau|\theta) = \log P(s_0) + \sum_{t=0}^{T} \log \pi_\theta(a_t|s_t) + \sum_{t=0}^{T} \log P(s_{t+1}|s_t, a_t)$$

Next, we compute the gradient of this log-probability expression with respect to θ. Because the initial state distribution $P(s_0)$ and the environment transition probabilities $P(s_{t+1}|s_t, a_t)$ are dictated by the environment and do not depend on the agent's policy parameters θ, their respective gradients vanish to zero:

$$\begin{aligned}\nabla_\theta \log P(\tau|\theta) &= \nabla_\theta \log P(s_0) + \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) + \sum_{t=0}^{T} \nabla_\theta \log P(s_{t+1}|s_t, a_t) \\ &= 0 + \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) + 0 = \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t)\end{aligned}$$

Finally, substituting this result back into the expectation expression completes the proof of the score function estimator:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau) \right]$$

Question 2: The Model-Free Property [2 Points]

In most reinforcement learning scenarios, the environment's transition dynamics $P(s_{t+1}|s_t, a_t)$ are completely unknown to the agent. Prove why we can still compute the policy gradient $\nabla_\theta \log P(\tau|\theta)$ from sample trajectories without needing to model or know the environment's transition dynamics. Show exactly what happens to the $P(s_{t+1}|s_t, a_t)$ term during the derivation.

Answer:

As established in the derivation of Question 1, the true policy gradient is expressed as an expectation over trajectories:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau) \right]$$

During the expansion of $\nabla_\theta \log P(\tau|\theta)$ in Question 1, the environment transition dynamics term $P(s_{t+1}|s_t, a_t)$ explicitly vanishes. Because the transition dynamics are dictated solely by the environment, they possess no mathematical dependence on the agent's policy parameters θ. Consequently, applying the gradient operator yields:

$$\nabla_\theta \log P(s_{t+1}|s_t, a_t) = 0$$

This causes the unknown transition probabilities to completely drop out of the final score function expression.

Since the final gradient formula depends only on the policy's own terms $\nabla_\theta \log \pi_\theta(a_t|s_t)$ and the experienced trajectory returns $R(\tau)$, we do not need an analytical model of the environment. By executing the current policy to collect M empirical sample trajectories from the environment, we can invoke the **Law of Large Numbers (LLN)**. The LLN guarantees that the empirical sample mean (Monte Carlo estimator):

$$\nabla_\theta J(\theta) \approx \frac{1}{M} \sum_{i=1}^{M} \left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t^{(i)}|s_t^{(i)}) R(\tau^{(i)}) \right]$$

unbiasedly converges to the true expected policy gradient as $M \to \infty$. This eliminates the need for modeling environment dynamics, proving the model-free property.

Question 3: Zero-Bias Baseline [2 Points]

Policy gradients suffer from high variance. To reduce this variance, we subtract a state-dependent function $b(s_t)$, known as a baseline, from our return. Prove that subtracting an arbitrary baseline $b(s_t)$ that depends only on the current state does not introduce any bias into the policy gradient estimate.

Answer:

To prove that subtracting a state-dependent baseline $b(s_t)$ introduces zero bias, we must demonstrate that the expected value of the baseline gradient term over the entire trajectory distribution is identically zero:

$$\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right] = 0$$

We proceed by breaking this down using three fundamental properties of probability and expectation:

1. Linearity of Expectation The horizon T is a fixed finite value for the given MDP formulation. By the **linearity of expectation**, the expectation of a sum is equal to the sum of the individual expectations, regardless of any dependencies between the random variables. This allows us to interchange the summation and the expectation operator:

$$\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right] = \sum_{t=0}^{T}\mathbb{E}_{\tau\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right]$$

2. Marginalization of the Trajectory Distribution Consider the expectation at a single fixed time step t. The objective function inside the expectation, $f(s_t, a_t) = \nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)$, depends strictly on the state s_t and action a_t at that specific time step. Expanding the full trajectory expectation yields:

$$\mathbb{E}_{\tau\sim\pi_\theta}\left[f(s_t, a_t)\right] = \sum_{s_0,a_0,\dots,s_T,a_T} P(s_0, a_0, \dots, s_T, a_T|\theta)f(s_t, a_t)$$

We factor the joint trajectory probability into three segments: the past (up to s_t), the current step (a_t), and the future (beyond a_t):

$$P(\tau|\theta) = \left(P(s_0)\prod_{k=0}^{t-1}\pi_\theta(a_k|s_k)P(s_{k+1}|s_k, a_k)\right)\cdot \pi_\theta(a_t|s_t)\cdot \left(\prod_{j=t}^{T-1}P(s_{j+1}|s_j, a_j)\pi_\theta(a_{j+1}|s_{j+1})\right)$$

When we sum over all possible future variables $(s_{t+1}, a_{t+1}, \dots, s_T, a_T)$, they sequentially collapse to 1 because they represent conditional probability distributions summed over their entire sample spaces (e.g., $\sum_{a_T}\pi_\theta(a_T|s_T) = 1$ and $\sum_{s_T} P(s_T|s_{T-1}, a_{T-1}) = 1$).
Simultaneously, summing the past variables $(s_0, a_0, \dots, s_{t-1}, a_{t-1})$ evaluates the marginal probability of arriving at state s_t at time step t, denoted as $P(s_t|\theta)$. Thus, the full trajectory probability marginalizes directly to the joint probability of the state-action pair at time t:

$$\mathbb{E}_{\tau\sim\pi_\theta}\left[f(s_t, a_t)\right] = \sum_{s_t,a_t} P(s_t, a_t|\theta)f(s_t, a_t) = \mathbb{E}_{(s_t,a_t)\sim P(s_t,a_t|\theta)}\left[f(s_t, a_t)\right]$$

3. Law of Total Expectation (The Tower Property) Using the relation $P(s_t, a_t|\theta) = P(s_t|\theta)\pi_\theta(a_t|s_t)$, we apply the **Law of Total Expectation** to decompose the joint expectation into an outer expectation over the state distribution and an inner conditional expectation over the action distribution given that state:

$$\mathbb{E}_{(s_t,a_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)\right] = \mathbb{E}_{s_t\sim P(s_t|\theta)}\left[\mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t) \mid s_t\right]\right]$$

We isolate and expand the inner conditional expectation over the action space $\mathcal{A}$ for any fixed state s_t:

$$\mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t) \mid s_t\right] = \sum_{a_t\in\mathcal{A}} \pi_\theta(a_t|s_t)\nabla_\theta \log \pi_\theta(a_t|s_t)b(s_t)$$

Using the log-derivative identity $\pi_\theta(a_t|s_t)\nabla_\theta \log \pi_\theta(a_t|s_t) = \nabla_\theta\pi_\theta(a_t|s_t)$, we substitute it into the sum:

$$= \sum_{a_t\in\mathcal{A}} \nabla_\theta\pi_\theta(a_t|s_t)b(s_t)$$

Because the baseline $b(s_t)$ is strictly state-dependent and acts as a constant with respect to the action summation, it can be factored out. Moreover, due to the linearity of the gradient operator ∇_θ, it commutes outside the summation:

$$= b(s_t)\sum_{a_t\in\mathcal{A}} \nabla_\theta\pi_\theta(a_t|s_t) = b(s_t)\nabla_\theta\left(\sum_{a_t\in\mathcal{A}} \pi_\theta(a_t|s_t)\right)$$

By the total probability axiom, the sum of probabilities of all possible actions from any given state must equal 1 ($\sum_{a_t}\pi_\theta(a_t|s_t) = 1$). Since the gradient of a constant scalar is zero ($\nabla_\theta(1) = 0$), we obtain:

$$= b(s_t)\cdot\nabla_\theta(1) = b(s_t)\cdot 0 = 0$$

Since the inner conditional expectation evaluates to exactly 0 for every achievable state s_t, the outer expectation over the state distribution is trivially zero:

$$\mathbb{E}_{s_t}[0] = 0 \implies \sum_{t=0}^{T} 0 = 0$$

This demonstrates that subtracting an arbitrary state-dependent baseline yields an expected value of zero, formally verifying that it introduces no bias into the true expected policy gradient estimate.

Question 4: The Optimal Baseline [3 Points]

Derive the exact, optimal baseline $b^*(s_t)$ that strictly minimizes the variance of the policy gradient estimate.

Answer:

To derive the exact optimal baseline $b^*(s_t)$ that strictly minimizes the variance of the total trajectory policy gradient, we define the total gradient estimator $g(\tau)$ over a complete trajectory τ as the sum of gradients across all time steps:

$$g(\tau) = \sum_{k=0}^{T} \nabla_\theta \log \pi_\theta(a_k|s_k)(R(\tau) - b_k(s_k))$$

For notation brevity, let $\psi_k = \nabla_\theta \log \pi_\theta(a_k|s_k)$ denote the score function vector at time step k. The variance of this cumulative vector-valued estimator is:

$$\mathrm{Var}(g(\tau)) = \mathbb{E}_\tau \left[\|g(\tau)\|^2\right] - \|\mathbb{E}_\tau[g(\tau)]\|^2$$

As established in Question 3, the expected value of the baseline component vanishes identically ($\mathbb{E}_\tau[\sum_k \psi_k b_k(s_k)] = 0$), meaning $\mathbb{E}_\tau[g(\tau)]$ is mathematically independent of the baseline choice. Thus, the second term $\|\mathbb{E}_\tau[g(\tau)]\|^2$ acts as a constant relative to $b_k(s_k)$. Minimizing the total variance is therefore equivalent to minimizing the total second moment:

$$\min_{b(\cdot)} \mathcal{L} = \min_{b(\cdot)} \mathbb{E}_\tau \left[\left\| \sum_{k=0}^{T} \psi_k (R(\tau) - b_k(s_k)) \right\|^2 \right]$$

To isolate the optimal baseline $b^*(s)$ for a specific designated state s at a specific time step t, we evaluate the functional partial derivative of $\mathcal{L}$ with respect to $b_t(s)$. By applying the chain rule, the derivative operator only activates when the trajectory actively visits state s at time step t. We express this localized activation via the indicator function $\mathbb{I}(s_t = s)$:

$$\frac{\partial \mathcal{L}}{\partial b_t(s)} = \mathbb{E}_\tau \left[2 \left(\sum_{k=0}^{T} \psi_k (R(\tau) - b_k(s_k)) \right)^\top (-\psi_t \cdot \mathbb{I}(s_t = s)) \right] = 0$$

Eliminating the constant -2 and expanding the expectation over the trajectory distribution yields:

$$\sum_\tau P(\tau|\theta) \cdot \mathbb{I}(s_t = s) \cdot \left[\psi_t^\top \sum_{k=0}^{T} \psi_k (R(\tau) - b_k(s_k)) \right] = 0$$

The indicator function restricts the summation strictly to the subset of trajectories where $s_t = s$. Factoring the joint probability as $P(\tau|\theta) = P(s_t = s|\theta) P(\tau \mid s_t = s)$, and dividing both sides by the state-visitation scalar $P(s_t = s|\theta) > 0$, we rigorously transform the full expectation into a conditional expectation:

$$\mathbb{E}_{\tau|s_t} \left[\sum_{k=0}^{T} \psi_t^\top \psi_k (R(\tau) - b_k(s_k)) \right] = 0$$

We partition this summation into the current-step main term ($k = t$) and the remaining cross-terms ($k \neq t$):

$$\mathbb{E}_{\tau|s_t} \left[\|\psi_t\|^2 (R(\tau) - b_t(s_t)) \right] + \sum_{k \neq t} \mathbb{E}_{\tau|s_t} \left[\psi_t^\top \psi_k (R(\tau) - b_k(s_k)) \right] = 0$$

We now explicitly expand the trajectory expectation to prove that the cross-terms vanish for two distinct structural cases. By writing out the full summation over all trajectory variables and rearranging the order of operations, we isolate the specific action that zeroes out each term:

- **Future Terms ($k > t$):** We push the summation over the future action a_k to the innermost level. The joint trajectory probability includes the policy term $\pi_\theta(a_k|s_k)$. Past variables (like ψ_t), past rewards, and the baseline $b_k(s_k)$ do not depend on a_k. Thus, the innermost summation successfully isolates ψ_k. By applying the log-derivative identity, $\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s)\nabla_\theta log \pi_\theta(a|s)$, we can simplify the inner summations:

$$\dots \left(\sum_{a_k \in \mathcal{A}} \pi_\theta(a_k|s_k)\psi_k \right) \dots = \dots \left(\sum_{a_k \in \mathcal{A}} \nabla_\theta \pi_\theta(a_k|s_k) \right) \dots = \dots (\nabla_\theta(1)) \dots = 0$$

Because this innermost factor evaluates to zero, the entire nested summation for any $k > t$ collapses to zero.

- **Past Terms ($k < t$):** Similarly, we push the summation over the current action a_t to the innermost level. The past variables ψ_k, past rewards, and $b_k(s_k)$ are entirely independent of a_t. The innermost summation now isolates ψ_t:

$$\dots \left(\sum_{a_t \in \mathcal{A}} \pi_\theta(a_t|s_t)\psi_t \right) \dots = \dots \left(\sum_{a_t \in \mathcal{A}} \nabla_\theta \pi_\theta(a_t|s_t) \right) \dots = \dots (\nabla_\theta(1)) \dots = 0$$

Again, the zeroing of this inner summation nullifies the entire outer expectation for any $k < t$.

Because the summation $\sum_a \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) = 0$ holds inherently at any isolated step, all covariance cross-terms elegantly decouple and evaluate to zero through direct algebraic marginalization.

Thus, the total variance minimization equation simplifies strictly to the individual component at time step t:

$$\mathbb{E}_{\tau|s_t} \left[\|\psi_t\|^2 (R(\tau) - b_t(s_t)) \right] = 0$$

Expanding the conditional expectation via its linear components yields:

$$\mathbb{E}_{\tau|s_t} \left[\|\psi_t\|^2 R(\tau) \right] - b_t(s_t)\mathbb{E}_{\tau|s_t} \left[\|\psi_t\|^2 \right] = 0$$

Solving explicitly for $b_t(s_t)$ defines the exact, variance-minimizing optimal baseline $b^*(s_t)$:

$$b^*(s_t) = \frac{\mathbb{E}_{\tau|s_t} \left[\|\nabla_\theta \log \pi_\theta(a_t|s_t)\|^2 R(\tau) \right]}{\mathbb{E}_{\tau|s_t} \left[\|\nabla_\theta \log \pi_\theta(a_t|s_t)\|^2 \right]}$$

This formally completes the derivation of the optimal baseline under the full trajectory reward formulation.

2 REINFORCE

Implementation 1: REINFORCE Implementation **[25 Points]**

1. REINFORCE without the causality trick, using the same full-episode return for every action.
2. REINFORCE with the causality trick, using reward-to-go.
3. REINFORCE with reward-to-go and a constant baseline.
4. REINFORCE with reward-to-go and a learned value-function baseline.

Answer:

See HW2_REINFORCE.ipynb.

Question 1: **[3 Points]**

Answer the questions inside the Jupyter notebook based on your experiments.

Answer:

See HW2_REINFORCE.ipynb.

Algorithm 1: Vanilla REINFORCE

Input: policy π_θ, learning rate α, discount factor γ, number of episodes N
for episode $= 1, \ldots, N$ do
    Roll out one trajectory using $a_t \sim \pi_\theta(\cdot|s_t)$
    Store rewards R_t and log-probabilities $\log \pi_\theta(a_t|s_t)$
    Compute the full discounted return $G_0 = \sum_{k=0}^{T} \gamma^k R_k$
    Form the loss $L(\theta) = -\sum_{t=0}^{T} \log \pi_\theta(a_t|s_t) G_0$
    Update θ by gradient descent on $L(\theta)$
end for

3 Policy-Induced Distributions and Gradients

3.1 Occupancy Measures

Question 1: Markov Chain as a Special Case [2 Points]

Consider a finite-horizon Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R)$ with horizon H, where the state space is partitioned into layers $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \cdots \cup \mathcal{S}_H$ and $\mathcal{S}_1 = \{s_{\text{init}}\}$ represents the initial state. Suppose $|\mathcal{A}| = 1$ (only one action is available per state).

- Explain how the MDP reduces to a Markov Chain (MC).
- In this MC, what does the occupancy measure q_π represent?
- If the MC were ergodic and we considered an infinite horizon ($H \to \infty$), how would the occupancy measure relate to the long-run behavior of the MC?

Answer:

We evaluate the reduction of the given layered MDP under the single-action constraint step-by-step:

1. Reduction of MDP to a Markov Chain When $|\mathcal{A}| = 1$, the action space collapses to a single unique action a for every state $s \in \mathcal{S}$. Consequently, the stochastic policy becomes trivial and deterministic, with $\pi(a|s) = 1$. The state transition probability function $P(s' \mid s, a)$ loses its dependency on an active control input and simplifies directly to a state-to-state transition probability:

$$P(s' \mid s, a) = P(s' \mid s)$$

Since the transition dynamics depend solely on the current state and are completely decoupled from any decision-making agent, the autonomous stochastic process strictly reduces to a standard Markov Chain (MC).

2. Structural Representation of the Occupancy Measure q_π Because the action space contains only a single option, the action dependency is eliminated, reducing $q_\pi(s, a)$ to a state-visitation probability $q_\pi(s)$. Given the layered partition of the state space $\mathcal{S}$, time step h is implicitly embedded within the state's identity (i.e., $s \in \mathcal{S}_h$ uniquely maps to time step h).
Crucially, $q_\pi(s)$ does not represent a cumulative hitting probability over the entire horizon; rather, it represents the **marginal probability** that the system occupies state s precisely at time step h. Because the agent must occupy exactly one state within the active layer $\mathcal{S}_h$ at step h, the occupancy measure satisfies a strict layer-wise normalization constraint:

$$\sum_{s \in \mathcal{S}_h} q_\pi(s) = 1 \quad \forall h \in \{1, \ldots, H\}$$

For any state $s \notin \mathcal{S}_h$ at time step h, its occupancy probability under this temporal slice is identically zero.

3. Infinite Horizon Limit ($H \to \infty$) under Ergodicity An ergodic Markov Chain possesses strong mixing properties, allowing the system to completely forget its initial condition s_{init} over an extended timeline. When the horizon approaches infinity ($H \to \infty$), the explicit temporal layering transitions into a continuous ongoing process.

Instead of collapsing to 1 (as a cumulative visitation probability would), the occupancy measure $q_\pi(s)$ converges smoothly to a stable, time-invariant probability vector. In this long-run equilibrium, $q_\pi(s)$ represents the exact **proportion of time** the process spends in state s. By the law of total probability, this limiting distribution forms a fixed-point equilibrium that satisfies the global balance equation:

$$q_\pi(s') = \sum_{s\in\mathcal{S}} q_\pi(s)P(s' \mid s) \implies q_\pi^\top P = q_\pi^\top$$

subject to the global normalization constraint across the entire state space: $\sum_{s\in\mathcal{S}} q_\pi(s) = 1$.

Question 2: Properties of the Occupancy Measure [2 Points]

Show that for any policy π, the induced occupancy measure q_π satisfies the following linear constraints:

$$\sum_{a\in A} q_\pi(s_{init}, a) = 1 \quad (1)$$

for each $h = 1, ..., H-1$ and each $s' \in S_{h+1}$:

$$\sum_{s\in S_h}\sum_{a\in A} q_\pi(s,a)P(s'|s,a) = \sum_{a\in A} q_\pi(s',a). \quad (2)$$

(Interpretation: Equation (2) is a flow conservation condition: the probability of entering state s equals the probability of being in s.)

Answer:

Proof of Equation (1): Initial Boundary Condition

By definition, the occupancy measure $q_\pi(s,a)$ represents the joint probability of visiting state s and selecting action a at the implicit time step h dictated by the layered partition $s \in \mathcal{S}_h$. For the initial step of the horizon ($h = 1$), the state space is a singleton containing exclusively the initial state, $\mathcal{S}_1 = \{s_{\text{init}}\}$.

Since the environment initializes deterministically at this state, the initial state distribution satisfies $P(S_1 = s_{\text{init}}) = 1$. Summing the occupancy measure over the entire action space $\mathcal{A}$ at this initial state yields the marginal probability of occupying s_{init} at $h = 1$:

$$\sum_{a\in\mathcal{A}} q_\pi(s_{\text{init}}, a) = \sum_{a\in\mathcal{A}} P(S_1 = s_{\text{init}}, A_1 = a)$$

Applying the law of total probability to marginalize out the action space $\mathcal{A}$ collapses the summation directly into the marginal probability of the initial state boundary:

$$= P(S_1 = s_{\text{init}}) = 1$$

This rigorously satisfies the initial condition stated in Equation (1).

Proof of Equation (2): Flow Conservation Condition

For any given time step $h \in \{1, \ldots, H-1\}$ and a designated successor state $s' \in \mathcal{S}_{h+1}$, the right-hand side of the equation characterizes the total marginal probability that the system occupies state s' at step $h+1$:

$$\sum_{a\in\mathcal{A}} q_\pi(s', a) = P(S_{h+1} = s')$$

To evaluate the left-hand side, we expand the occupancy measure into its joint probability representation for occupying state $s \in \mathcal{S}_h$ and executing action $a \in \mathcal{A}$ at time step h:

$$\sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s,a) P(s' \mid s,a) = \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} P(S_h = s, A_h = a) P(S_{h+1} = s' \mid S_h = s, A_h = a)$$

By the definition of conditional probability, the product of the prior joint probability and the transition dynamics yields the full joint probability of the state-action-state sequence:

$$= \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} P(S_h = s, A_h = a, S_{h+1} = s')$$

By invoke the law of total probability to sequentially marginalize out all possible actions $a \in \mathcal{A}$ and all achievable predecessor states $s \in \mathcal{S}_h$ from the previous temporal layer, the nested summation simplifies directly to the marginal probability of occupying state s' at step $h+1$:

$$= P(S_{h+1} = s')$$

Since both sides of the equation evaluate identically to $P(S_{h+1} = s')$, the flow conservation constraint is formally proved.

Question 3: From Constraints to Occupancy Measure [4 Points]

Now suppose we have a function $q : \mathcal{S} \times \mathcal{A} \to [0,1]$ that satisfies constraints (1) and (2). Define a policy π by:

$$\pi(a|s) = \frac{q(s,a)}{\sum_{a' \in \mathcal{A}} q(s,a')} \quad \text{if} \quad \sum_{a' \in \mathcal{A}} q(s,a') > 0 \quad (3)$$

and let $\pi(\cdot|s)$ be an arbitrary distribution over $\mathcal{A}$ otherwise. Prove that q is the occupancy measure induced by π, i.e., $q = q_\pi$.

Answer:

We prove that $q(s,a) = q_\pi(s,a)$ for all state-action pairs across the layered state space $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \cdots \cup \mathcal{S}_H$ using mathematical induction directly on the layer index h.
For brevity, we denote the marginal state-visitation probability as $q(s) \equiv \sum_{a \in \mathcal{A}} q(s,a)$ and $q_\pi(s) \equiv \sum_{a \in \mathcal{A}} q_\pi(s,a)$. Under the explicit condition $q(s) > 0$ given in the prompt, the policy definition in Equation (3) implies $q(s,a) = \pi(a|s)q(s)$. Since the true induced occupancy measure structurally satisfies $q_\pi(s,a) = \pi(a|s)q_\pi(s)$, proving $q(s) = q_\pi(s)$ immediately guarantees $q(s,a) = q_\pi(s,a)$.
Base Case: Layer $h = 1$
The initial layer contains only the singleton initial state, $\mathcal{S}_1 = \{s_{\text{init}}\}$. According to Constraint (1), the denominator for this initial state evaluates to exactly 1:

$$q(s_{\text{init}}) = \sum_{a \in \mathcal{A}} q(s_{\text{init}}, a) = 1$$

By the definition of the true occupancy measure, the system initializes at s_{init} with absolute certainty, meaning $q_\pi(s_{\text{init}}) = 1$. Since $q(s_{\text{init}}) = q_\pi(s_{\text{init}}) = 1 > 0$, we substitute this into the policy relation to yield:

$$q(s_{\text{init}}, a) = \pi(a|s_{\text{init}})q(s_{\text{init}}) = \pi(a|s_{\text{init}}) \cdot 1 = \pi(a|s_{\text{init}})q_\pi(s_{\text{init}}) = q_\pi(s_{\text{init}}, a)$$

This satisfies the base case for the first layer.

Inductive Hypothesis

Assume that the identity holds true for all temporal layers from 1 up to h:

$$q(s,a) = q_\pi(s,a) \quad \text{and} \quad q(s) = q_\pi(s) \quad \forall s \in \{\mathcal{S}_1, \dots, \mathcal{S}_h\}, \forall a \in \mathcal{A}$$

Inductive Step: Layer $h+1$

We evaluate an arbitrary successor state $s' \in \mathcal{S}_{h+1}$ where $q(s') > 0$. By utilizing the flow conservation property specified in Constraint (2), we express $q(s')$ as a function of the previous layer $\mathcal{S}_h$:

$$q(s') = \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q(s,a) P(s' \mid s,a)$$

Applying the inductive hypothesis to replace the predecessor terms $q(s,a)$ with $q_\pi(s,a)$ yields:

$$q(s') = \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s,a) P(s' \mid s,a)$$

Because the true induced occupancy measure q_π inherently satisfies this exact same flow conservation identity (as proven in Question 2), the right-hand side simplifies directly to the true marginal state distribution at step $h+1$:

$$q(s') = q_\pi(s')$$

Since both denominators are identical and strictly positive ($q(s') = q_\pi(s') > 0$), we can directly combine this result with the definition of the policy $\pi(a|s')$ from Equation (3):

$$q(s',a) = \pi(a|s')q(s') = \pi(a|s')q_\pi(s') = q_\pi(s',a)$$

This completes the inductive step for all reachable states in layer $h+1$, formally proving that $q = q_\pi$ over the entire horizon.

Question 4: Utility of the Occupancy Measure Formulation [1 Points]

- Express the expected total loss $v_\pi(s_{\text{init}})$ in terms of q_π and γ.
- Discuss one advantage and one disadvantage of optimizing over the space of occupancy measures (subject to constraints (1) and (2)) compared to optimizing directly over the policy space.

Answer:

We provide the analytical formulation and the conceptual trade-offs of the occupancy measure framework below.

1. Analytical Expression for Expected Total Loss

Given the layered structure of the state space $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \dots \cup \mathcal{S}_H$, the occupancy measure $q_\pi(s,a)$ represents the marginal probability of visiting the state-action pair (s,a) at the specific time step h corresponding to layer $\mathcal{S}_h$. By leveraging the linearity of expectation, the expected total discounted loss $v_\pi(s_{\text{init}})$ can be expressed as the weighted sum of the per-step cost $R(s,a)$ multiplied by its probability of occurrence and the temporal discount factor γ^{h-1}:

$$v_\pi(s_{\text{init}}) = \sum_{h=1}^{H} \gamma^{h-1} \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s,a) R(s,a)$$

2. Optimization Trade-offs: Advantage and Disadvantage
Advantage: Linearity and Convexity (Global Optimality)
When optimizing directly over the policy space π, the objective function is highly non-convex due to the complex, non-linear compounding interactions of the policy across sequential time steps. Conversely, in the occupancy measure space, both the objective function $v_\pi(s_{\text{init}})$ derived above and the flow conservation constraints (1) and (2) are strictly **linear** with respect to the decision variables $q(s,a)$. This completely transforms a challenging non-convex optimization problem into a standard Linear Program (LP). Consequently, it guarantees convergence to the absolute global optimum in polynomial time using robust solvers (e.g., simplex or interior-point methods), entirely bypassing the issues of local minima, saddle points, or vanishing gradients.
Disadvantage: Curse of Dimensionality and Numerical Instability
The primary disadvantage is the curse of dimensionality; the number of optimization variables scales directly with the total cardinality of the state-action space ($|\mathcal{S}| \times |\mathcal{A}|$), which becomes mathematically and computationally intractable for large discrete spaces and impossible to represent tabularly for continuous environments. Additionally, executing the optimal solution requires extracting the policy via the relation $\pi(a|s) = q(s,a)/\sum_{a'} q(s,a')$. For unvisited or low-probability states where the total visitation fluid approaches zero ($\sum_{a'} q(s,a') \approx 0$), this division introduces extreme numerical instability, resulting in undefined or arbitrary action distributions that fail to generalize robustly.

3.2 Discounted Visitation and the Gradient of v_π

Question 5: The Gradient of v_π **[5 Points]**

Prove Equation 3 holds for all $s \in \mathcal{S}$ in a discounted MDP.

Answer:

textbfAnalytical Note on Notation Alignment
We explicitly note a severe notation clash in the assignment text between Section 3.1 and Section 3.2. While $q_\pi(s,a)$ was defined as the probability-valued occupancy measure in Section 3.1 , the formulation of the policy gradient in Equation 3 strictly requires $q_\pi(s,a)$ to act as the standard **action-value function** (or Q-function), which structurally encapsulates the environment's reward dynamics. Following standard reinforcement learning conventions established later in Section 3.3, we treat $q_\pi(s,a)$ here as the expected discounted cumulative return:

$$q_\pi(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) v_\pi(s')$$

Neumann Series Expansion and Physical Interpretation of $Pr_\pi(s'|s)$
Let $P_\pi \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ denote the state-to-state transition probability matrix induced under the current policy π_θ. Each element of this matrix represents the marginalized probability of transitioning from

state s to state s' in exactly one step:

$$[P_\pi]_{ss'} = \sum_{a \in \mathcal{A}} \pi_\theta(a \mid s) P(s' \mid s, a)$$

The operator defined in Equation 4, $Pr_\pi(s'|s) := [(I_n - \gamma P_\pi)^{-1 Mosley}]_{ss'}$, can be rigorously expanded as an infinite matrix geometric series (the Neumann series), since the discount factor satisfies $\gamma < 1$:

$$(I_n - \gamma P_\pi)^{-1} = I_n + \gamma P_\pi + \gamma^2 P_\pi^2 + \gamma^3 P_\pi^3 + \cdots = \sum_{t=0}^{\infty} \gamma^t P_\pi^t$$

Physically, the matrix power P_π^t dictates the state-to-state transition probabilities over exactly t time steps. Taking the specific entry at row s and column s' yields:

$$Pr_\pi(s' \mid s) = \sum_{t=0}^{\infty} \gamma^t P(S_t = s' \mid S_0 = s)$$

This value represents the expected discounted total number of time steps the agent spends occupying state s' over an infinite horizon, given that the trajectory initiated at state s.

Derivation of the Policy Gradient Equation

We begin by taking the functional gradient of the state-value function $v_\pi(s)$ with respect to the policy parameters θ. Expressing the state-value as the expectation of the action-value function over the policy yields:

$$v_\pi(s) = \sum_{a \in \mathcal{A}} \pi_\theta(a \mid s) q_\pi(s, a)$$

Applying the product rule of calculus to compute the gradient $\nabla_\theta v_\pi(s)$, we obtain:

$$\nabla_\theta v_\pi(s) = \sum_{a \in \mathcal{A}} [\nabla_\theta \pi_\theta(a \mid s) q_\pi(s, a) + \pi_\theta(a \mid s) \nabla_\theta q_\pi(s, a)]$$

Crucially, by the model-free property of this framework, the transition dynamics $P(s' \mid s, a)$ and the immediate reward function $R(s, a)$ are fixed properties of the environment and are entirely independent of the policy parameters θ. Therefore, the gradient operator distributes exclusively to the parameterized terms inside $q_\pi(s, a)$:

$$\nabla_\theta q_\pi(s, a) = \nabla_\theta \left(R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) v_\pi(s') \right) = \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \nabla_\theta v_\pi(s')$$

Substituting this back into our primary gradient equation yields:

$$\nabla_\theta v_\pi(s) = \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) q_\pi(s, a) + \gamma \sum_{a \in \mathcal{A}} \pi_\theta(a \mid s) \sum_{s' \in \mathcal{S}} P(s' \mid s, a) \nabla_\theta v_\pi(s')$$

By changing the order of the summations in the second component, we isolate the state-to-state transition matrix component $[P_\pi]_{ss'}$:

$$\nabla_\theta v_\pi(s) = \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) q_\pi(s, a) + \gamma \sum_{s' \in \mathcal{S}} \left(\sum_{a \in \mathcal{A}} \pi_\theta(a \mid s) P(s' \mid s, a) \right) \nabla_\theta v_\pi(s')$$

$$\nabla_\theta v_\pi(s) = \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) q_\pi(s, a) + \gamma \sum_{s' \in \mathcal{S}} [P_\pi]_{ss'} \nabla_\theta v_\pi(s')$$

To solve this recursive relation, we define $\nabla_\theta v_\pi \in \mathbb{R}^{|\mathcal{S}|\times m}$ as a vector/matrix over the state space, and let $\phi \in \mathbb{R}^{|\mathcal{S}|\times m}$ denote the local immediate gradient vector whose elements are $\phi(s) = \sum_{a\in\mathcal{A}} \nabla_\theta \pi_\theta(a \mid s) q_\pi(s,a)$. Writing the equation in compact matrix notation gives:

$$\nabla_\theta v_\pi = \phi + \gamma P_\pi \nabla_\theta v_\pi$$

Grouping the policy gradient terms onto the left-hand side yields:

$$(I_n - \gamma P_\pi)\nabla_\theta v_\pi = \phi \implies \nabla_\theta v_\pi = (I_n - \gamma P_\pi)^{-1}\phi$$

Finally, by extracting the specific scalar equation for a designated state s, and substituting the Neumann series entry representation for $[(I_n - \gamma P_\pi)^{-1}]_{ss'} = Pr_\pi(s'|s)$, we arrive at:

$$\nabla_\theta v_\pi(s) = \sum_{s'\in\mathcal{S}} Pr_\pi(s' \mid s)\phi(s') = \sum_{s'\in\mathcal{S}} Pr_\pi(s' \mid s) \sum_{a\in\mathcal{A}} \nabla_\theta \pi_\theta(a \mid s') q_\pi(s', a)$$

This matches Equation 3 identically for all $s \in \mathcal{S}$, completing the formal proof.

3.3 Undiscounted Gradients

Lemma 1 – Equivalence of Average Reward Rate

Let d_π be the stationary probability distribution vector satisfying the transition equilibrium $d_\pi^\top P_\pi = d_\pi^\top$ and the standard normalization constraint $d_\pi^\top 1_n = 1$. For the long-run average reward rate $\overline{r}_\pi$, the following identity holds strictly:

$$d_\pi^\top r_\pi = \overline{r}_\pi$$

Proof: By the algebraic properties of an undiscounted infinite-horizon MDP, there exists a differential state-value vector v_π that satisfies the steady-state Poisson equation: $v_\pi = r_\pi - \overline{r}_\pi 1_n + P_\pi v_\pi$. To utilize this equilibrium, we multiply the entire Poisson matrix relation from the left by the stationary distribution row vector $d_\pi^\top$:

$$d_\pi^\top v_\pi = d_\pi^\top (r_\pi - \overline{r}_\pi 1_n + P_\pi v_\pi)$$

Distributing the row vector across the linear terms on the right-hand side yields:

$$d_\pi^\top v_\pi = d_\pi^\top r_\pi - \overline{r}_\pi (d_\pi^\top 1_n) + d_\pi^\top P_\pi v_\pi$$

Substituting the given defining properties of the stationary distribution, namely $d_\pi^\top P_\pi = d_\pi^\top$ and $d_\pi^\top 1_n = 1$, the equation simplifies directly to:

$$d_\pi^\top v_\pi = d_\pi^\top r_\pi - \overline{r}_\pi(1) + d_\pi^\top v_\pi$$

Since $d_\pi^\top v_\pi$ is a scalar, subtracting it from both sides of the equality isolates the intended relation:

$$0 = d_\pi^\top r_\pi - \overline{r}_\pi \implies d_\pi^\top r_\pi = \overline{r}_\pi$$

This concludes the formal proof of the lemma.

Question 6: Solution of the Poisson Equation [5 Points]

Show that

$$v_\pi^* = (I_n - P_\pi + 1_n d_\pi^\top)^{-1} r_\pi \quad (7)$$

is a solution of the Poisson equation . Here, the distribution d_π is selected such that it is stationary distribution under π, satisfying $d_\pi^\top P_\pi = d_\pi^\top$.

Answer:

We formally verify that the proposed vector v_π^* satisfies the Poisson equation by leveraging the invariant relationship established in Lemma 1.

Proof of Equation (7)

We begin by multiplying both sides of the proposed solution in Equation (7) from the left by the regularized matrix operator $(I_n - P_\pi + 1_n d_\pi^\top)$ to eliminate the matrix inverse :

$$(I_n - P_\pi + 1_n d_\pi^\top) v_\pi^* = r_\pi$$

Expanding the operators across the state-value vector v_π^* yields the following linear relation:

$$v_\pi^* - P_\pi v_\pi^* + 1_n (d_\pi^\top v_\pi^*) = r_\pi \quad (\star)$$

To evaluate the scalar component $d_\pi^\top v_\pi^*$, we project the entire vector relation $(\star)$ by multiplying it from the left by the stationary distribution row vector $d_\pi^\top$:

$$d_\pi^\top v_\pi^* - d_\pi^\top P_\pi v_\pi^* + d_\pi^\top 1_n (d_\pi^\top v_\pi^*) = d_\pi^\top r_\pi$$

Applying the core stationary distribution property $d_\pi^\top P_\pi = d_\pi^\top$ simplifies the difference of the first two terms to zero ($d_\pi^\top v_\pi^* - d_\pi^\top v_\pi^* = 0$). Substituting the standard distribution normalization condition $d_\pi^\top 1_n = 1$ reduces the equation to:

$$1 \cdot (d_\pi^\top v_\pi^*) = d_\pi^\top r_\pi \implies d_\pi^\top v_\pi^* = d_\pi^\top r_\pi$$

By direct invocation of Lemma 1, the term $d_\pi^\top r_\pi$ is identically equal to the long-run average reward rate $\bar{r}_\pi$. Therefore, we establish that $d_\pi^\top v_\pi^* = \bar{r}_\pi$. Finally, substituting this scalar identity back into our expanded relation $(\star)$ yields:

$$v_\pi^* - P_\pi v_\pi^* + \bar{r}_\pi 1_n = r_\pi$$

Rearranging the vectors to isolate v_π^* on the left-hand side fully recovers the standard Poisson equation :

$$v_{\pi*} = r_\pi - \bar{r}_\pi 1_n + P_\pi v_\pi^*$$

This formally demonstrates that v_π^* satisfies the equilibrium condition, proving it is a valid solution.

Question 7: General Form of Solutions [6 Points]

Show that any solution of the Poisson equation has the following form:

$$v^\pi = v_\pi^* + c 1_n \quad (8)$$

where $c \in \mathbb{R}$.

Answer:

We establish the general algebraic structure of the solutions to the Poisson equation by examining the eigenspace of the state transition matrix.

1. Derivation of the Difference System

Let $v^\pi \in \mathbb{R}^n$ be an arbitrary vector that satisfies the Poisson equation :

$$v^\pi = r_\pi - \overline{r}_\pi 1_n + P_\pi v^\pi$$

From Question 6, we know that v^*_π is also a valid particular solution to the exact same system :

$$v^*_\pi = r_\pi - \overline{r}_\pi 1_n + P_\pi v^*_\pi$$

Subtracting the second equation from the first eliminates both the baseline immediate reward vector r_π and the constant average reward shift vector $\overline{r}_\pi 1_n$:

$$v^\pi - v^*_\pi = P_\pi v^\pi - P_\pi v^*_\pi = P_\pi(v^\pi - v^*_\pi)$$

By defining the difference vector as $u = v^\pi - v^*_\pi$, this relationship simplifies directly to a standard homogeneous linear system / eigenvalue problem:

$$u = P_\pi u \implies (I_n - P_\pi)u = 0$$

This demonstrates that the difference vector u must reside in the nullspace (kernel) of the matrix $(I_n - P_\pi)$, which is mathematically equivalent to saying that u is a right eigenvector of P_π corresponding to the eigenvalue $\lambda = 1$.

2. Eigenspace Analysis via Row-Stochasticity and Ergodicity

To uniquely characterize the form of u, we analyze the spectral properties of the transition operator P_π:

- **Row-Stochastic Property:** By construction, P_π is a row-stochastic matrix because each row represents a valid conditional probability distribution over the next state space , strictly satisfying $\sum_{j=1}^{n} [P_\pi]_{ij} = 1$ for all rows i. Multiplying P_π from the right by the all-ones column vector $1_n = [1, 1, \ldots, 1]^\top$ sums the elements of each row, which yields:

  $$P_\pi 1_n = 1_n \implies (I_n - P_\pi)1_n = 0$$

  This identities that the all-ones vector 1_n is guaranteed to be a right eigenvector of P_π for the eigenvalue $\lambda = 1$, meaning $1_n \in \text{null}(I_n - P_\pi)$.

- **Ergodicity and Nullspace Simplicity:** Under the explicit assumption that the policy π induces an ergodic Markov chain with a single recurrent class , the Perron-Frobenius theorem dictates that the maximum eigenvalue of P_π is strictly simple. This guarantees that the algebraic and geometric multiplicity of the eigenvalue $\lambda = 1$ is exactly equal to 1.

3. Completing the General Form Representation

Because the geometric multiplicity of $\lambda = 1$ is exactly 1, the nullspace $\text{null}(I_n - P_\pi)$ is strictly a one-dimensional vector subspace. A one-dimensional subspace is fully spanned by any single non-zero vector residing within it. Since we established that the non-zero vector 1_n belongs to this nullspace, it forms a complete and sufficient basis for the kernel:

$$\text{null}(I_n - P_\pi) = \text{span}(1_n)$$

Consequently, any difference vector u that satisfies the homogeneous condition $(I_n - P_\pi)u = 0$ has no choice but to be a scalar multiple of the basis vector 1_n. This implies there exists a constant $c \in \mathbb{R}$ such that:

$$u = c1_n$$

Substituting the definition $u = v^\pi - v_\pi^*$ back into this relation yields:

$$v^\pi - v_\pi^* = c1_n \implies v^\pi = v_\pi^* + c1_n$$

This completes the formal proof that any alternative valid solution to the Poisson equation can only differ from the particular solution v_π^* by a uniform constant shift across all states.

4 Natural Policy Gradient (NPG)

4.1 Average-Reward Policy Optimization

In continuing tasks, the average-reward objective $\eta(\theta)$ can be formulated as a long-run time average or via the stationary state distribution $\rho^\pi(s)$:

$$\eta(\theta) = \lim_{T\to\infty} \frac{1}{T}\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} R(s_t, a_t)\right] = \sum_{s\in S} \rho^\pi(s) \sum_{a\in A} \pi_\theta(a|s) R(s, a)$$

The ultimate optimization goal is to find the policy parameters θ that maximize this long-run average reward rate: $\max_\theta \eta(\theta)$.

4.2 The Differential Action-Value Function

Question 1: Finiteness of the Differential Action-Value Function **[4 Points]**

Prove that, under the assumptions that S and A are finite, rewards are bounded, and the induced Markov chain under the fixed policy π is ergodic, the differential action-value function

$$Q^\pi(s, a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty}(R(s_t, a_t) - \eta(\pi))\middle| s_0 = s, a_0 = a\right]$$

is finite for every state-action pair (s, a).

Answer:

To prove that $Q^\pi(s, a)$ remains finite for all $(s, a) \in S \times A$, we analyze the convergence of its defining infinite series by leveraging the geometric mixing property of finite ergodic Markov chains.

1. Fundamental Bounds and Definitions

- **Bounded Rewards:** Since the reward function $R(s, a)$ is bounded over the finite state-action spaces, there exists a uniform upper bound $R_{\max} < \infty$ such that:

$$\forall s \in S, a \in A : \quad |R(s, a)| \leq R_{\max}$$

- **Expected State Reward:** Let $r_\pi(s) = \sum_{a\in A} \pi(a|s)R(s, a)$ denote the expected immediate reward in state s under policy π. By linearity, it inherits the same bound: $|r_\pi(s)| \leq R_{\max}$.

- **Average Reward Rate:** The long-run average reward per time step is defined as $\eta(\pi) = \sum_{s\in S} \rho^\pi(s) r_\pi(s)$, where ρ^π is the unique stationary distribution satisfying $\sum_{s\in S} \rho^\pi(s) = 1$. It follows directly that $|\eta(\pi)| \le R_{\max}$.

2. Geometric Mixing Property For any finite, ergodic Markov chain, the t-step transition probability distribution converges to the unique stationary distribution ρ^π at a geometric (exponential) rate. Specifically, there exist constants $C > 0$ and a contraction factor $\gamma \in [0, 1)$ such that for any state $s' \in S$:

$$\left|\mathbb{P}(s_t = s' \mid s_0 = s, a_0 = a) - \rho^\pi(s')\right| \le C\gamma^t$$

3. Bounding the Sequence Terms Let Δ_t denote the expected deviation of the reward from the long-run average rate at time step t:

$$\Delta_t = \mathbb{E}_\pi \left[R(s_t, a_t) - \eta(\pi) | s_0 = s, a_0 = a\right]$$

For any step $t \ge 1$, conditioning on the intermediate state $s_t = s'$ yields:

$$\begin{aligned}\Delta_t &= \sum_{s'\in S} \mathbb{P}(s_t = s' \mid s_0 = s, a_0 = a) r_\pi(s') - \sum_{s'\in S} \rho^\pi(s') r_\pi(s') \\ &= \sum_{s'\in S} \left(\mathbb{P}(s_t = s' \mid s_0 = s, a_0 = a) - \rho^\pi(s')\right) r_\pi(s')\end{aligned}$$

Applying the triangle inequality alongside the geometric mixing bound and the reward bound, we obtain:

$$|\Delta_t| \le \sum_{s'\in S} \left|\mathbb{P}(s_t = s' \mid s_0 = s, a_0 = a) - \rho^\pi(s')\right| \cdot |r_\pi(s')| \le \sum_{s'\in S} (C\gamma^t) R_{\max} = |S| C R_{\max} \gamma^t$$

4. Proof of Series Convergence We can now evaluate the absolute value of the total infinite series by partitioning it at an arbitrary finite time threshold $t_0 \ge 1$:

$$|Q^\pi(s, a)| \le \sum_{t=0}^{\infty} |\Delta_t| = \sum_{t=0}^{t_0-1} |\Delta_t| + \sum_{t=t_0}^{\infty} |\Delta_t|$$

1. **Transient Phase ($t < t_0$):** Each early term is trivially bounded via the triangle inequality by $|\Delta_t| \le |\mathbb{E}[R_t]| + |\eta(\pi)| \le 2R_{\max}$. Summing over the first t_0 steps yields:

$$\sum_{t=0}^{t_0-1} |\Delta_t| \le 2t_0 R_{\max}$$

2. **Asymptotic Phase ($t \ge t_0$):** For the remaining tail of the series, we apply our geometric mixing bound and evaluate the resulting infinite geometric progression:

$$\sum_{t=t_0}^{\infty} |\Delta_t| \le \sum_{t=t_0}^{\infty} |S| C R_{\max} \gamma^t = |S| C R_{\max} \frac{\gamma^{t_0}}{1-\gamma}$$

Combining both parts results in a well-defined global upper bound:

$$|Q^\pi(s, a)| \le 2t_0 R_{\max} + \frac{|S| C R_{\max} \gamma^{t_0}}{1-\gamma}$$

Since the state space is finite ($|S| < \infty$), the rewards are bounded ($R_{\max} < \infty$), and the contraction coefficient is strictly less than one ($\gamma < 1$), the right-hand side is a finite real number. Consequently, $|Q^\pi(s, a)| < \infty$, which proves that the differential action-value function is finite for every state-action pair (s, a).

Question 2: Advantage-Like Interpretation [2 Points]

The corresponding state-value function is

$$J^\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)}[Q^\pi(s,a)].$$

Show that replacing $Q^\pi(s,a)$ by

$$A^\pi(s,a) = Q^\pi(s,a) - J^\pi(s)$$

does not change the policy gradient.

Answer:

According to the average-reward policy-gradient theorem, the ordinary policy gradient of the long-run average reward objective can be expressed as an expectation over the stationary state distribution and the policy's action distribution:

$$\nabla\eta(\theta) = \mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s,a)\right]$$

Now, we substitute the differential advantage function $A^\pi(s,a) = Q^\pi(s,a) - J^\pi(s)$ in place of $Q^\pi(s,a)$ within the policy gradient expectation:

$$\nabla\eta_A(\theta) = \mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) \left(Q^\pi(s,a) - J^\pi(s)\right)\right]$$

By the linearity of expectation, this expression can be expanded into two distinct components:

$$\nabla\eta_A(\theta) = \mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s,a)\right] - \mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) J^\pi(s)\right]$$

To demonstrate that this substitution leaves the policy gradient completely unchanged ($\nabla\eta_A(\theta) = \nabla\eta(\theta)$), it is necessary and sufficient to prove that the expected value of the second term containing the state-value function $J^\pi(s)$ evaluates exactly to zero:

$$\mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) J^\pi(s)\right] = 0$$

This mathematically mirrors the zero-bias baseline property analyzed in Section 1 Question 3, where subtracting an arbitrary state-dependent baseline $b(s_t) = J^\pi(s_t)$ from the return objective introduces absolutely zero bias into the gradient estimator.

Proof that the Baseline Term Vanishes:

We expand the expectation of the baseline term explicitly into its structural summations over the finite state and action spaces:

$$\mathbb{E}_{s\sim\rho^\pi, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s) J^\pi(s)\right] = \sum_{s\in S} \rho^\pi(s) \sum_{a\in A} \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) J^\pi(s)$$

Using the fundamental log-derivative score identity $\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta \pi_\theta(a|s)$, we rewrite the inner sum:

$$= \sum_{s\in S} \rho^\pi(s) \sum_{a\in A} \nabla_\theta \pi_\theta(a|s) J^\pi(s)$$

Since the differential state-value function $J^\pi(s)$ depends strictly on the state s and carries no algebraic dependence on the individual action a, it acts as a constant relative to the inner summation and can be factored out cleanly:

$$= \sum_{s\in S} \rho^\pi(s) J^\pi(s) \sum_{a\in A} \nabla_\theta \pi_\theta(a|s)$$

Because the action space A is finite and its boundaries are independent of the parameter vector θ, the gradient operator can be moved outside the summation over actions:

$$= \sum_{s \in S} \rho^{\pi}(s) J^{\pi}(s) \nabla_{\theta} \left(\sum_{a \in A} \pi_{\theta}(a|s) \right)$$

By the axioms of probability, the sum of the policy's probabilities over the entire action simplex must equal exactly 1 for any given state s ($\sum_{a \in A} \pi_{\theta}(a|s) = 1$). Substituting this constant gives:

$$= \sum_{s \in S} \rho^{\pi}(s) J^{\pi}(s) \nabla_{\theta}(1)$$

Since the gradient of any constant scalar with respect to the parameter vector θ is identically zero ($\nabla_{\theta}(1) = 0$), the entire inner expression vanishes:

$$= \sum_{s \in S} \rho^{\pi}(s) J^{\pi}(s) \cdot \mathbf{0} = 0$$

Because this baseline expectation is zero, the subtractive term disappears completely, leaving the gradient identical to the original objective formulation. Thus, replacing $Q^{\pi}(s, a)$ with the advantage function $A^{\pi}(s, a)$ does not alter the policy gradient.

4.3 The Ordinary Policy Gradient

The ordinary policy gradient computes the average-reward objective gradient via

$$\nabla \eta(\theta) = \mathbb{E}_{s \sim \rho^{\pi}, a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a)]$$

. The standard update $\theta \leftarrow \theta + \alpha \nabla \eta(\theta)$ implicitly treats the parameter space as a flat Euclidean geometry where $\|\Delta\theta\|^2 = \Delta\theta^{\top}\Delta\theta$. However, since parameters are just a coordinate system, ordinary gradient steps depend on the specific parameterization rather than the actual change in the policy distribution.

4.4 The Fisher Geometry of Policies

Question 3: Natural Gradient as Steepest Ascent **[4 Points]**

Derive the natural-gradient direction by solving the local constrained optimization problem

$$\max_{\Delta\theta} \nabla \eta(\theta)^{\top} \Delta\theta$$

subject to

$$\Delta\theta^{\top} F(\theta) \Delta\theta = \epsilon.$$

Answer:

This problem can be rigorously solved as a classic constrained optimization problem using the method of Lagrange multipliers, extended to vector-valued variables .
For algebraic convenience, we simplify the notation by defining the ordinary policy gradient vector as $g \doteq \nabla \eta(\theta)$ and the Fisher information matrix as $F \doteq F(\theta)$. The optimization problem is

reframed as:

$$\max_{\Delta\theta} \quad g^\top \Delta\theta \quad \text{subject to} \quad \Delta\theta^\top F \Delta\theta = \epsilon$$

1. Constructing the Lagrangian
We introduce the Lagrange multiplier $\lambda \in \mathbb{R}$ to integrate the equality constraint into the objective function:

$$\mathcal{L}(\Delta\theta, \lambda) = g^\top \Delta\theta + \lambda \left(\epsilon - \Delta\theta^\top F \Delta\theta\right)$$

2. First-Order Necessary Condition
Taking the gradient of the Lagrangian $\mathcal{L}$ with respect to the parameter step vector $\Delta\theta$ and setting it to zero yields:

$$\nabla_{\Delta\theta}\mathcal{L}(\Delta\theta, \lambda) = g - 2\lambda F \Delta\theta = \mathbf{0}$$

Isolating the term involving the Fisher matrix gives:

$$2\lambda F \Delta\theta = g \implies F\Delta\theta = \frac{1}{2\lambda} g$$

Assuming the Fisher information matrix F is strictly positive-definite and thus invertible, we multiply from the left by F^{-1}:

$$\Delta\theta = \frac{1}{2\lambda} F^{-1} g$$

This condition reveals that the optimal step direction is directly proportional to the inverse Fisher-weighted gradient:

$$\Delta\theta \propto F^{-1} \nabla\eta(\theta)$$

which is the defined direction of the **Natural Policy Gradient** .
3. Solving for the Exact Closed-Form Step
To eliminate the unknown multiplier λ, we substitute our expression for $\Delta\theta$ back into the quadratic constraint boundary $\Delta\theta^\top F \Delta\theta = \epsilon$:

$$\left(\frac{1}{2\lambda} F^{-1} g\right)^\top F \left(\frac{1}{2\lambda} F^{-1} g\right) = \epsilon$$

Using the transpose property $(AB)^\top = B^\top A^\top$ and noting that F (and consequently F^{-1}) is symmetric, we expand the forms:

$$\frac{1}{4\lambda^2} g^\top F^{-1} F F^{-1} g = \epsilon \implies \frac{1}{4\lambda^2} g^\top F^{-1} g = \epsilon$$

Solving explicitly for the scaling parameter 2λ, we obtain:

$$4\lambda^2 = \frac{g^\top F^{-1} g}{\epsilon} \implies 2\lambda = \sqrt{\frac{g^\top F^{-1} g}{\epsilon}} \implies \frac{1}{2\lambda} = \sqrt{\frac{\epsilon}{g^\top F^{-1} g}}$$

Substituting this precise scalar value back into the directional step equation yields the exact closed-form solution for the parameter update:

$$\Delta\theta = \sqrt{\frac{\epsilon}{\nabla\eta(\theta)^\top F^{-1}(\theta) \nabla\eta(\theta)}} F^{-1}(\theta) \nabla\eta(\theta)$$

Intuitive Interpretation: While the ordinary gradient vector identifies the steepest path along a flat Euclidean coordinate space, the Fisher matrix tracks the underlying curvature of the actual policy distribution space. Pre-multiplying by F^{-1} stretches or compresses the step sizes to normalize for this curvature, ensuring the policy distribution morphs smoothly by an exact, controlled magnitude ϵ regardless of the parameterization chosen.

4.5 Compatible Function Approximation

Question 4: Compatible Function Approximation and Natural Gradient [3 Points]

Assume w minimizes

$$\epsilon(w,\pi) = \sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)(w^\top \psi^\pi(s,a) - Q^\pi(s,a))^2$$

Show that $w = F(\theta)^{-1}\nabla\eta(\theta)$.

Answer:

To find the vector w that minimizes the error $\epsilon(w,\pi)$, we compute its gradient with respect to w and set it to zero:

$$\nabla_w \epsilon(w,\pi) = 2\sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\psi^\pi(s,a)\left(w^\top \psi^\pi(s,a) - Q^\pi(s,a)\right) = \mathbf{0}$$

Expanding the terms and noting that $w^\top\psi^\pi(s,a) = \psi^\pi(s,a)^\top w$ is a scalar, we isolate w:

$$\left[\sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\psi^\pi(s,a)\psi^\pi(s,a)^\top\right] w = \sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\psi^\pi(s,a)Q^\pi(s,a)$$

By definition, the score feature is $\psi^\pi(s,a) = \nabla_\theta \log \pi_\theta(a|s)$. Substituting this into the equation:

- **Left-Hand Side(LHS):** Matches the exact definition of the Fisher information matrix $F(\theta)$:

$$\sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s)^\top = F(\theta)$$

- **Right-Hand Side(RHS):** Utilizing the identity $\pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta \pi_\theta(a|s)$, it simplifies to the ordinary policy gradient $\nabla\eta(\theta)$:

$$\sum_{s,a} \rho^\pi(s)\nabla_\theta \pi_\theta(a|s)Q^\pi(s,a) = \nabla\eta(\theta)$$

Substituting these definitions back into the linear system yields:

$$F(\theta)w = \nabla\eta(\theta)$$

Assuming $F(\theta)$ is invertible, we left-multiply by $F(\theta)^{-1}$ to get the final closed-form solution:

$$w = F(\theta)^{-1}\nabla\eta(\theta)$$

4.6 Why the Compatible Approximation is Advantage-Like

Question 5: Zero-Mean Property **[2 Points]**

Prove that

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[f^\pi(s,a;w)] = 0$$

Answer:

To prove the zero-mean property of the compatible function approximator, consider:

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[f^\pi(s,a;w)] = \sum_{a\in A} \pi_\theta(a|s) f^\pi(s,a;w)$$

Substituting $f^\pi(s,a;w) = w^\top \psi^\pi(s,a)$ and $\psi^\pi(s,a) = \nabla_\theta \log \pi_\theta(a|s)$:

$$\begin{aligned}
\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[f^\pi(s,a;w)] &= \sum_{a\in A} \pi_\theta(a|s) \Big(w^\top \nabla_\theta \log \pi_\theta(a|s) \Big) \\
&= w^\top \sum_{a\in A} \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \\
&= w^\top \sum_{a\in A} \nabla_\theta \pi_\theta(a|s) \\
&= w^\top \nabla_\theta \left(\sum_{a\in A} \pi_\theta(a|s) \right) \\
&= w^\top \nabla_\theta (1) \\
&= w^\top \mathbf{0} = 0
\end{aligned}$$

Here we used the identity

$$\pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta \pi_\theta(a|s)$$

together with the normalization condition

$$\sum_{a\in A} \pi_\theta(a|s) = 1.$$

Hence the expectation of $f^\pi(s,a;w)$ under the policy is zero for every state s. Therefore, the compatible function approximator naturally behaves as an advantage function rather than an absolute action-value function.

4.7 Natural Gradient and Greedy Policy Improvement

Question 6: Computing the Score Feature **[4 Points]**

Considering the exponential-family policy, show that the score feature is

$$\psi^\pi(s,a) = \phi_{s,a} - \mathbb{E}_{b\sim\pi_\theta(\cdot|s)}[\phi_{s,b}].$$

Answer:

By definition, the score feature $\psi^\pi(s,a)$ is the gradient of the log-policy:

$$\psi^\pi(s,a) = \nabla_\theta \log \pi_\theta(a|s)$$

Given the exponential-family (softmax) policy $\pi_\theta(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{\sum_b \exp(\theta^\top \phi_{s,b})}$, we first apply the log-division property ($\log \frac{A}{B} = \log A - \log B$):

$$\log \pi_\theta(a|s) = \theta^\top \phi_{s,a} - \log \sum_b \exp(\theta^\top \phi_{s,b})$$

Now, we compute the gradient with respect to θ by applying the linearity of the gradient and the chain rule ($\nabla \log U = \frac{\nabla U}{U}$):

$$\nabla_\theta \log \pi_\theta(a|s) = \nabla_\theta(\theta^\top \phi_{s,a}) - \frac{\nabla_\theta \sum_b \exp(\theta^\top \phi_{s,b})}{\sum_b \exp(\theta^\top \phi_{s,b})}$$

Evaluating the derivatives yields:

$$= \phi_{s,a} - \frac{\sum_b \exp(\theta^\top \phi_{s,b}) \cdot \phi_{s,b}}{\sum_b \exp(\theta^\top \phi_{s,b})}$$

Distributing the denominator inside the summation reveals the policy term $\pi_\theta(b|s)$:

$$= \phi_{s,a} - \sum_b \left(\frac{\exp(\theta^\top \phi_{s,b})}{\sum_c \exp(\theta^\top \phi_{s,c})} \right) \phi_{s,b} = \phi_{s,a} - \sum_b \pi_\theta(b|s) \phi_{s,b}$$

By definition, the summation over the policy probabilities represents the expectation of the features over actions:

$$\psi^\pi(s,a) = \phi_{s,a} - \mathbb{E}_{b\sim\pi_\theta(\cdot|s)}[\phi_{s,b}]$$

Question 7: Large-Step Greedy Limit [4 Points]

Define $\pi_\infty(a|s) = \lim_{\alpha\to\infty} \pi_{\theta+\alpha\tilde{\nabla}\eta(\theta)}(a|s)$. Show that $\pi_\infty(a|s) \neq 0$ if and only if $a \in \arg\max_{a'} f^\pi(s,a';w)$.

Answer:

To evaluate the large-step limit of the natural policy gradient update, we analyze the behavior of the updated exponential-family policy as the step-size parameter α approaches infinity .

1. Expressing the Policy Update via $f^\pi(s,a;w)$

From the result of Question 4, the natural policy gradient direction corresponds exactly to the optimal compatible weight vector, $\tilde{\nabla}\eta(\theta) = w$. Therefore, the parameter update is $\theta' = \theta + \alpha w$. Substituting θ' into the exponential-family policy definition gives:

$$\pi_{\theta+\alpha w}(a|s) = \frac{\exp\left((\theta+\alpha w)^\top \phi_{s,a}\right)}{\sum_b \exp\left((\theta+\alpha w)^\top \phi_{s,b}\right)} = \frac{\exp\left(\theta^\top \phi_{s,a} + \alpha w^\top \phi_{s,a}\right)}{\sum_b \exp\left(\theta^\top \phi_{s,b} + \alpha w^\top \phi_{s,b}\right)}$$

Using the score feature identity from Question 6, we know that $\psi^\pi(s,a) = \phi_{s,a} - \mathbb{E}_{b\sim\pi_\theta}[\phi_{s,b}]$. Linear multiplication by $w^\top$ yields the compatible function approximation form:

$$f^\pi(s,a;w) = w^\top\psi^\pi(s,a) = w^\top\phi_{s,a} - w^\top\mathbb{E}_{b\sim\pi_\theta}[\phi_{s,b}] \implies w^\top\phi_{s,a} = f^\pi(s,a;w) + C(s)$$

where $C(s) = \mathbb{E}_{b\sim\pi_\theta}[w^\top\phi_{s,b}]$ is a baseline term independent of individual actions. Substituting $w^\top\phi_{s,a}$ back into the policy k-th exponential component allows $\exp(\alpha C(s))$ to factor out and cancel completely from both the numerator and denominator:

$$\pi_{\theta+\alpha\tilde{\nabla}\eta(\theta)}(a|s) = \frac{\exp\left(\theta^\top\phi_{s,a} + \alpha f^\pi(s,a;w)\right)}{\sum_b \exp\left(\theta^\top\phi_{s,b} + \alpha f^\pi(s,b;w)\right)}$$

2. Evaluating the Asymptotic Limit ($\alpha \to \infty$)
Let $f^*(s) = \max_{a'} f^\pi(s,a';w)$ represent the maximum value attainable by the compatible function approximation at state s. We divide every term in the softmax fraction by $\exp(\alpha f^*(s))$:

$$\pi_{\theta+\alpha\tilde{\nabla}\eta(\theta)}(a|s) = \frac{\exp\left(\theta^\top\phi_{s,a}\right)\exp\left(\alpha\left[f^\pi(s,a;w) - f^*(s)\right]\right)}{\sum_b \exp\left(\theta^\top\phi_{s,b}\right)\exp\left(\alpha\left[f^\pi(s,b;w) - f^*(s)\right]\right)}$$

Taking the limit as $\alpha \to \infty$, the term $\alpha\left[f^\pi(s,a;w) - f^*(s)\right]$ governs the exponential scaling:

- If $a \notin \arg\max_{a'} f^\pi(s,a';w)$, then $f^\pi(s,a;w) < f^*(s)$, meaning the exponent is strictly negative. As $\alpha \to \infty$, $\exp\left(\alpha\left[f^\pi(s,a;w) - f^*(s)\right]\right) \to 0$.
- If $a \in \arg\max_{a'} f^\pi(s,a';w)$, then $f^\pi(s,a;w) = f^*(s)$, meaning the exponent is exactly zero. Thus, $\exp(0) = 1$ regardless of the value of α.

Evaluating the limit equation explicitly yields the closed-form definition for $\pi_\infty(a|s)$:

$$\pi_\infty(a|s) = \begin{cases} \frac{\exp\left(\theta^\top\phi_{s,a}\right)}{\sum_{b\in\arg\max_{a'} f^\pi(s,a';w)} \exp\left(\theta^\top\phi_{s,b}\right)} & \text{if } a \in \arg\max_{a'} f^\pi(s,a';w) \\ 0 & \text{if } a \notin \arg\max_{a'} f^\pi(s,a';w) \end{cases}$$

Since $\exp\left(\theta^\top\phi_{s,a}\right) > 0$ for all valid bounded feature mappings, the numerator of the active case is strictly positive. Consequently, the limiting distribution probability satisfies $\pi_\infty(a|s) \neq 0$ if and only if $a \in \arg\max_{a'} f^\pi(s,a';w)$. This completes the proof, verifying that an infinite natural policy gradient step converts the stochastic policy into a greedy improvement step over the compatible value function.

4.8 Local Policy Improvement for General Policies

Question 8: Local Natural-Gradient Update **[4 Points]**

For a general smooth policy, prove the local first-order relationship between $\pi_{\theta'}(a|s)$ and $\pi_\theta(a|s)$ under the update $\theta' = \theta + \alpha\tilde{\nabla}\eta(\theta)$. Then, show that for sufficiently small α, to first order, actions with larger values of $f^\pi(s,a;w)$ receive larger multiplicative increases in probability.

Answer:

To analyze the local behavior of a general smooth policy under a small natural policy gradient update, we derive its first-order relationship and inspect its multiplicative properties.

1. Derivation of the Local First-Order Relationship

Given the parameter update rule $\theta' = \theta + \alpha\tilde{\nabla}\eta(\theta)$, where α is a sufficiently small step size, we can write the local first-order linear approximation of the policy function $\pi_{\theta'}(a|s)$ around the current parameters θ as follows:

$$\pi_{\theta'}(a|s) \approx \pi_\theta(a|s) + \alpha\nabla_\theta\pi_\theta(a|s)^\top\tilde{\nabla}\eta(\theta)$$

We simplify this expression by substituting two established identities from our framework:

- From the log-derivative identity, the standard policy gradient vector relates to the score feature via: $\nabla_\theta\pi_\theta(a|s) = \pi_\theta(a|s)\nabla_\theta \log\pi_\theta(a|s) = \pi_\theta(a|s)\psi^\pi(s,a)$.
- From Question 4, the natural policy gradient direction corresponds exactly to the optimal compatible function approximation weight vector: $\tilde{\nabla}\eta(\theta) = w$.

Substituting these components back into the linear approximation yields:

$$\pi_{\theta'}(a|s) \approx \pi_\theta(a|s) + \alpha\left(\pi_\theta(a|s)\psi^\pi(s,a)\right)^\top w \implies \pi_{\theta'}(a|s) \approx \pi_\theta(a|s) + \alpha\pi_\theta(a|s)\left(w^\top\psi^\pi(s,a)\right)$$

By definition, the inner product $w^\top\psi^\pi(s,a)$ is exactly the compatible function approximation $f^\pi(s,a;w)$. Replacing this term gives:

$$\pi_{\theta'}(a|s) \approx \pi_\theta(a|s) + \alpha\pi_\theta(a|s)f^\pi(s,a;w)$$

Factoring out the original policy probability $\pi_\theta(a|s)$, we arrive at the final local first-order relationship:

$$\pi_{\theta'}(a|s) \approx \pi_\theta(a|s)\left(1 + \alpha f^\pi(s,a;w)\right)$$

2. Multiplicative Increase Interpretation

To show the multiplicative growth property, we isolate the relative change in the action probability by dividing both sides of the derived update equation by $\pi_\theta(a|s)$:

$$\frac{\pi_{\theta'}(a|s)}{\pi_\theta(a|s)} \approx 1 + \alpha f^\pi(s,a;w)$$

The ratio $\frac{\pi_{\theta'}(a|s)}{\pi_\theta(a|s)}$ represents the scaling factor applied to the original probability. For a fixed, positive, and sufficiently small step size $\alpha > 0$, the multiplicative growth factor is strictly determined by the value of $f^\pi(s,a;w)$.

Consequently, actions that yield a higher value under the compatible function approximation $f^\pi(s,a;w)$ will directly receive a larger multiplicative coefficient. This mathematically proves that a local natural policy gradient step dynamically shifts the policy distribution to amplify the selection probability of the most advantageous actions.

4.9 Why This is Not Newton's Method

Question 9: Conceptual Comparison [2 Points]

Explain why the natural gradient is not the same as Newton's method. What does the Fisher matrix measure, and what additional quantities appear in the Hessian of the RL objective (write the additional terms)?

Answer:

The natural policy gradient method is fundamentally distinct from Newton's optimization method due to the underlying spaces their respective curvature matrices evaluate .

1. What Each Matrix Measures

- **Newton's Method (Hessian Matrix):** Measures the geometric curvature of the **objective function landscape** ($\eta(\theta)$) over the raw parameter space. It dictates how the objective value accelerates or decelerates with respect to θ.
- **Natural Gradient (Fisher Matrix):** Measures the internal curvature of the **policy distribution space** (π_θ). It computes the statistical sensitivity (local KL divergence) of the action probabilities without regarding the reward values directly.

2. The Hessian of the RL Objective and Additional Terms

The ordinary policy gradient is defined as $\nabla\eta(\theta) = \sum_{s,a} \rho^\pi(s)\nabla_\theta \pi_\theta(a|s)Q^\pi(s,a)$. Taking the second derivative with respect to θ yields the full Hessian matrix $\nabla^2\eta(\theta)$:

$$\nabla^2\eta(\theta) = \sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\left[\nabla_\theta \log \pi_\theta(a|s)\nabla_\theta \log \pi_\theta(a|s)^\top\right] Q^\pi(s,a) + \text{Additional Terms}$$

While the first component resembles a Q-weighted variation of the Fisher Information Matrix $F(\theta) = \mathbb{E}[\nabla \log \pi \nabla \log \pi^\top]$, the actual Hessian contains several **additional terms** obtained via the product rule:

$$\text{Additional Terms} = \sum_{s,a} \rho^\pi(s)\pi_\theta(a|s)\nabla^2_\theta \log \pi_\theta(a|s)Q^\pi(s,a)$$

$$+\sum_{s,a}\left(\nabla_\theta\rho^\pi(s)\nabla_\theta\pi_\theta(a|s)^\top Q^\pi(s,a) + \rho^\pi(s)\nabla_\theta\pi_\theta(a|s)\nabla_\theta Q^\pi(s,a)^\top\right)$$

Where the explicit additional quantities that appear are:

1. $\nabla^2_\theta \log \pi_\theta(a|s)Q^\pi(s,a)$: The second-order derivative of the log-policy scaled by the action values.
2. $\nabla_\theta\rho^\pi(s)$: The gradient of the stationary state distribution, capturing how parameter changes alter state-visitation frequencies.
3. $\nabla_\theta Q^\pi(s,a)$: The gradient of the differential action-value function, capturing how parameter updates influence future cumulative reward structures.

Because the Fisher matrix omits the second derivative of the policy, changes in state dynamics ($\nabla\rho^\pi$), and changes in future rewards (∇Q^π), it is not equivalent to the objective Hessian.

4.10 Natural Gradient as a Small Policy-Distribution Step

Question 10: KL Divergence and the Fisher Matrix **[2 Points]**

Write the Taylor expansion of $\overline{D}_{KL}(\theta, \theta + \Delta\theta)$, around $\Delta\theta = 0$ up to second order. Then explain why the natural-gradient direction is the steepest ascent direction under a local KL constraint.

Answer:

To evaluate the local second-order curvature of the policy change, we compute the Taylor expansion of the expected KL divergence $f(\Delta\theta) = \overline{D}_{KL}(\theta, \theta + \Delta\theta)$ around $\Delta\theta = \mathbf{0}$:

$$\overline{D}_{KL}(\theta, \theta + \Delta\theta) \approx \overline{D}_{KL}(\theta, \theta) + \nabla_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}}^{\top} \Delta\theta + \frac{1}{2}\Delta\theta^{\top}\left[\nabla^2_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}}\right]\Delta\theta$$

1. Zeroth and First-Order Terms

By definition, the KL divergence of any distribution with respect to itself is identically zero, establishing the zeroth-order term:

$$\overline{D}_{KL}(\theta, \theta) = 0$$

Next, expanding the expectation form of the average KL divergence :

$$\overline{D}_{KL}(\theta, \theta + \Delta\theta) = \mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\pi_{\theta}(a|s)\log\pi_{\theta}(a|s) - \sum_{a\in A}\pi_{\theta}(a|s)\log\pi_{\theta+\Delta\theta}(a|s)\right]$$

Taking the first partial derivative with respect to $\Delta\theta$, only the second term depends on the perturbation vector:

$$\nabla_{\Delta\theta}\overline{D}_{KL}(\theta, \theta+\Delta\theta) = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\pi_{\theta}(a|s)\nabla_{\Delta\theta}\log\pi_{\theta+\Delta\theta}(a|s)\right] = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\frac{\pi_{\theta}(a|s)}{\pi_{\theta+\Delta\theta}(a|s)}\nabla_{\theta}\pi_{\theta+\Delta\theta}(a|s)\right]$$

Evaluating this gradient at the point of origin $\Delta\theta = \mathbf{0}$ yields:

$$\nabla_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}} = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\nabla_{\theta}\pi_{\theta}(a|s)\right] = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\nabla_{\theta}\left(\sum_{a\in A}\pi_{\theta}(a|s)\right)\right]$$

Because the probabilities sum to a constant ($\sum_a \pi_{\theta}(a|s) = 1$), its gradient vanishes completely:

$$\nabla_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}} = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\nabla_{\theta}(1)\right] = \mathbf{0}$$

2. Second-Order Curvature Term (Hessian)

Differentiating the gradient expression a second time with respect to $\Delta\theta$ via the quotient rule or standard log-Hessian identity gives:

$$\nabla^2_{\Delta\theta}\overline{D}_{KL}(\theta, \theta + \Delta\theta) = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\pi_{\theta}(a|s)\left(\frac{\nabla^2_{\theta}\pi_{\theta+\Delta\theta}(a|s)}{\pi_{\theta+\Delta\theta}(a|s)} - \frac{\nabla_{\theta}\pi_{\theta+\Delta\theta}(a|s)\nabla_{\theta}\pi_{\theta+\Delta\theta}(a|s)^{\top}}{\pi_{\theta+\Delta\theta}(a|s)^2}\right)\right]$$

Evaluating the full matrix equation explicitly at the boundary line $\Delta\theta = \mathbf{0}$:

$$\nabla^2_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}} = -\mathbb{E}_{s\sim\rho^{\pi}}\left[\nabla^2_{\theta}\sum_{a\in A}\pi_{\theta}(a|s)\right] + \mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\pi_{\theta}(a|s)\frac{\nabla_{\theta}\pi_{\theta}(a|s)\nabla_{\theta}\pi_{\theta}(a|s)^{\top}}{\pi_{\theta}(a|s)^2}\right]$$

The first term vanishes again because $\nabla^2_{\theta}(1) = \mathbf{0}$. Utilizing the identity $\frac{\nabla_{\theta}\pi_{\theta}(a|s)}{\pi_{\theta}(a|s)} = \nabla_{\theta}\log\pi_{\theta}(a|s)$, the expression simplifies directly to:

$$\nabla^2_{\Delta\theta}\overline{D}_{KL}(\theta, \theta)\Big|_{\Delta\theta=\mathbf{0}} = \mathbb{E}_{s\sim\rho^{\pi}}\left[\sum_{a\in A}\pi_{\theta}(a|s)\nabla_{\theta}\log\pi_{\theta}(a|s)\nabla_{\theta}\log\pi_{\theta}(a|s)^{\top}\right] = F(\theta)$$

which is the exact formulation of the expected Fisher information matrix.

3. Final Taylor Form and Geometric Interpretation

Substituting the evaluated terms back into the local quadratic system yields the clean closed-form expansion:

$$\overline{D}_{KL}(\theta, \theta + \Delta\theta) \approx \frac{1}{2}\Delta\theta^\top F(\theta)\Delta\theta$$

This proof implies that the Fisher matrix serves as the local quadratic approximation of the KL divergence between adjacent policies.

When finding the directional update that maximizes the first-order local reward improvement $\max_{\Delta\theta} \nabla\eta(\theta)^\top\Delta\theta$ subject to keeping the expected policy distribution change bounded within a small constant step ($\overline{D}_{KL}(\theta, \theta + \Delta\theta) = \epsilon$), the optimization becomes constrained exactly by $\frac{1}{2}\Delta\theta^\top F(\theta)\Delta\theta = \epsilon$. Solving this via Lagrange multipliers yields $\Delta\theta \propto F(\theta)^{-1}\nabla\eta(\theta)$, proving that the natural policy gradient is the optimal steepest ascent direction under a localized KL-divergence constraint.

Question 11: PG vs. NPG [1 Points]

In the notebook we compare ordinary policy gradient and natural policy gradient under both fixed-learning-rate updates and KL-scaled updates. Explain whether the observed behavior agrees with the theoretical interpretation developed in this section.

Answer:

The empirical results observed in both experiments perfectly align with the theoretical frameworks developed throughout this section.

Under fixed learning-rate updates, the Natural Policy Gradient (NPG) exhibits a drastically faster convergence rate compared to the standard Policy Gradient (PG), achieving optimal performance with a batch average reward of approximately 0.4 and a mean of 120 lines cleared within the first 100 iterations, whereas PG requires nearly 500 to 600 iterations to reach a similar level. This behavior perfectly matches the theoretical alignment because standard PG optimizes under a Euclidean parameter metric ($\Delta\theta^\top\Delta\theta = \epsilon$), causing it to take dangerously large or inefficiently small steps in the actual policy distribution space depending on the specific parameter coordinates. NPG corrects this coordinate dependency by utilizing the Fisher Information Matrix $F(\theta)$ to enforce a uniform step size directly in the probability distribution space ($\Delta\theta^\top F(\theta)\Delta\theta = \epsilon$), which guarantees high sample efficiency and rapid, monotonic policy improvement.

Furthermore, when the step sizes are adaptively scaled using a target KL divergence constraint, the performance gap between PG and NPG narrows significantly as the ordinary PG experiences a massive boost in convergence speed, learning almost as rapidly as NPG while NPG retains a slight edge in asymptotic stability. This empirical outcome validates the proof from Question 10, which established that the Fisher Information Matrix serves as the local second-order Taylor approximation of the expected KL divergence ($\overline{D}_{KL} \approx \frac{1}{2}\Delta\theta^\top F(\theta)\Delta\theta$). By manually forcing standard PG to bound its updates within a localized KL-divergence envelope, we artificially replicate the distribution-aware step regulation that NPG naturally computes via its metric tensor, thereby preventing destructive steps in distribution space and dramatically stabilizing the standard policy gradient algorithm.